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Merging Modules

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Part I

Design and commentary

1 Background

[bg]

1.1 Introduction

[bg.intro]

At the Jacksonville 2018 committee meeting, P0947R0 (“Another Take On Modules”, hereafter referred to as *Atom*) was presented. Two options were polled:

- merging the Atom proposal with the Modules TS, and
- progressing the Atom proposal as a separate TS.

Both options passed, but the first option had stronger support.

At Rapperswill 2018, we presented P1103R0, representing our effort in merging the two proposals and the remaining outstanding questions from the merge. Many open questions were answered, and we voted to adopt the merged proposal into the Modules TS, and scheduled a 2-day ad-hoc meeting to discuss the remaining open questions.

At the Bellevue 2018 ad-hoc Modules meeting, we answered all remaining known open design questions. This paper provides a description for the resulting design, as well as wording (based on the wording in P1103R0, which in turn is based on the wording of the Modules TS) to incorporate this design into the C++20 working draft.

1.2 Acknowledgements

[bg.ack]

This paper is based on the work of a great many people. The author would like to thank them all, and specifically Gabriel Dos Reis, on whose Modules TS this paper is primarily based.

2 Summary of merged proposal [merged]

2.1 Basics [merged.basic]

- ¹ A *module unit* is a translation unit that forms part of a module. Such a translation unit begins with¹ a preamble, comprising a module declaration and a sequence of imports:

```
exportopt module foo;
import a;
export import b;
// ... more imports ...
```

Within a module unit, imports may not appear after the end of the preamble. The **export** keyword indicates that a module unit is a *module interface unit*, which defines the interface for the module. For a module `foo`, there must be exactly one translation unit whose preamble contains **export module foo**;. This is the *primary module interface unit* for `foo` (2.2). (All other module interface units are module interface partitions; see 2.2.)

- ² Two related but distinct notions are key to understanding module semantics:

- (2.1) — A declaration is *visible* in a context if it can be found by a suitable name lookup.
- (2.2) — A declaration is *reachable* in a context if its semantic effects are available for use. (For example, a class type is complete in contexts where a definition of the class is reachable.)

A declaration is reachable wherever it is visible, but the converse is not true in general. Imports control which namespace-scope names are visible to name lookup, and which declarations are reachable semantically. The behavior of an entity is determined by the set of reachable declarations of that entity. For example, class members and enumeration members are visible to name lookup if there is a reachable definition of the class or enumeration.

- ³ A declaration can be exported by use of the **export** keyword in a module interface unit:

```
export int a;
export {
void f();
}
```

Exported declarations in a module interface unit are visible to name lookup in contexts that import that module interface unit. Non-exported declarations (excluding those with internal linkage) in a module unit are visible to name lookup in contexts within the same module that import the module unit. All declarations in transitively-imported module units are reachable, whether or not they are exported. [Example:

```
// a.cpp
export module A; // interface of module A
int foo() { return 1; }
export int bar();

// a-impl.cpp
module A;           // implementation of module A, implicitly
                    // considered to import the interface of A
```

1) A global module fragment (2.3.1) may optionally precede the preamble.

```

// OK; foo() is visible here (in the same module) even though
// it was not exported from A
int bar() { return foo() + 1; }

// unrelated.cpp
import A;
int main() {
    bar();           // OK, bar was exported by A and is visible here
    foo();           // error: foo is not visible here
}

```

— end example]

- ⁴ If a declaration within a namespace is exported, the enclosing namespace is also implicitly exported (but other declarations in the namespace are not implicitly exported). If a namespace is explicitly exported, all declarations within that namespace definition are exported.

2.2 Module partitions

[merged.part]

- ¹ A complete module can be defined in a single source file. However, the design, nature, and size of a module may warrant dividing both the implementation and the interface into multiple files. Module partitions provide facilities to support this.
- ² The module interface may be split across multiple files, if desired. Such files are called *module interface partitions*, and are introduced by a module declaration containing a colon:

```
export module foo:part;
```

- ³ Module interface partitions behave logically like distinct modules, except that they share ownership of contained entities with the module that they form part of. This allows an entity to be declared in one partition and defined in another, which may be necessary to resolve dependency cycles. It also permits code to be moved between partitions of a module with no impact on ABI.
- ⁴ The primary module interface unit for a module is required to transitively import and re-export all of the interface partitions of the module.
- ⁵ When the implementation of a module is split across multiple files, it may be desirable to share declarations between the implementation units without including them in the module interface unit, in order to avoid all consumers of the module having a physical dependency on the implementation details. (Specifically, if the implementation details change, the module interface and its dependencies should not need to be rebuilt.) This is made possible by *module implementation partitions*, which are module partitions that do not form part of the module interface:

```
module foo:part;
```

- ⁶ Module implementation partitions cannot contain exported declarations; instead, all declarations within them are visible to other translation units in the same module that import the partition. [*Note*: Exportation only affects which names and declarations are visible outside the module. — end note]
- ⁷ Module implementation partitions can be imported into the interface of a module, but cannot be exported.
- ⁸ Module interface partitions and module implementation partitions are collectively known as *module partitions*. Module partitions are an implementation detail of the module, and cannot be named outside the module. To emphasize this, an import declaration naming a module partition cannot be given a module name, only a partition name:

```

module foo;
import :part;           // imports foo:part
import bar:part;        // syntax error

```

```
import foo:part;           // syntax error
```

2.3 Support for non-modular code [merged.nonmodular]

- ¹ This proposal provides several features to support interoperability between modular code and traditional non-modular code.

2.3.1 Global module fragment [merged.global.frag]

- ¹ The merged proposal permits Modules TS-style global module fragments, with the `module;` introducer proposed in P0713R1 and approved by EWG:

```
module;
#include "some-header.h"
export module foo;
// ... use declarations and macros from some-header.h ...
```

- ² Prior to preprocessing, only preprocessor directives can appear in the global module fragment, but those directives can include `#include` directives that expand to declarations, as usual. Declarations in the global module fragment are not owned by the module.
- ³ In order to avoid bloating the interface of a module with declarations included into its global module fragment, declarations in the global module fragment that are not (transitively) referenced by the module unit are discarded. In particular, such declarations are not reachable from other translation units that import the module unit, and cannot be found by the second phase of two-phase name lookup for a template instantiation whose point of instantiation is outside the module unit. A declaration in a global module fragment is considered to be referenced if it is named within the module unit (after the preamble), or if it is mentioned by a referenced declaration.

- ⁴ [Example:

```
module;
#include "some-header.h" // defines classes X, Y, Z
export module foo;

export using X = ::X; // export name X; retain all declarations
                      // of X from "some-header.h"
export Y f(); // export name f; retain all declarations
              // of Y from "some-header.h"
// Z is not mentioned, so is discarded
```

— end example]

2.3.2 Header units [merged.header.import]

- ¹ The merged proposal also permits Atom-style header units, which are introduced by a special `import` syntax that names a header file instead of a module:

```
export module foo;
import "some-header.h";
import <version>;
// ... use declarations and macros from some-header.h and <version> ...
```

- ² The named header is processed as if it was a source file, the interface of the header is extracted and made available for import, and any macros defined by preprocessing the header are saved so that they can be made available to importers.

- ³ Declarations from code in a header unit are not owned by any module. In particular, the same entities can be redeclared by another header unit or by non-modular code. Header units can be re-exported using the regular `export import` syntax:

```
export module foo;
export import "some-header.h";
```

However, when a header unit is re-exported, macros are not exported. Only the header import syntax can import macros.

2.3.3 Reachability of nonmodular declarations [merged.nonmodular.reachability]

- ¹ A declaration in a global module fragment or header unit is reachable if it is visible. It is unspecified whether such a declaration is also reachable in contexts where it is not visible but is transitively imported. (Ideally, such a declaration would not be considered reachable in such contexts. However, in practice, making transitively-imported declarations unreachable would impose a severe implementation cost for some implementations, so we leave the extent to which this rule is enforced up to the implementation.)
- ² Discarded declarations (2.3.1) from the global module fragment are never visible nor reachable from outside that module unit.

2.3.4 Module use from non-modular code [merged.nonmodular.use]

- ¹ Modules and header units can be imported into non-modular code. Such imports can appear anywhere, and are not restricted to a preamble. This permits “bottom-up” modularization, whereby a library switches to providing only a modular interface and defining its header interface in terms of the modular interface. Non-modular code includes translation units other than module units (including headers imported as header units), and the global module fragment of a module unit.
- ² When a `#include` appears within non-modular code, if the named header file is known to correspond to a header unit, the implementation treats the `#include` as an import of the corresponding header unit. The mechanism for discovering this correspondence is left implementation-defined; there are multiple viable strategies here (such as explicitly building header units and providing them as input to downstream compilations, or introducing accompanying files describing the header unit structure) and we wish to encourage exploration of this space. An implementation is also permitted to not provide any mapping mechanism, and process each header unit independently.

3 Comparison to prior proposals [vs]

3.1 Changes since R2 [vs.r2]

¹ This section lists changes to the design of the merged modules proposal since P1103R2.

- (1.1) — Addressed wording feedback from CWG and other parties.
- (1.2) — Fixed bug where an incidentally-reachable definition could render other definitions of the same entity ill-formed:

```
// a.h
#ifndef A_H
#define A_H
class X {};
#endif

// b.cc
#include "a.h"
export module B;
X x;

// c.cc
export module C;
import B; // not exported

// d.cc
import C;
#include "a.h" // OK even if definition of X is incidentally visible
                // due to semantic boundaries rule
```

- (1.3) — Fixed unintended divergence from Modules TS that permitted module linkage entities to have definitions in multiple module units

3.2 Changes since R1 [vs.r1]

¹ This section lists changes to the design of the merged modules proposal since P1103R1.

3.2.1 P1299R2: Module preamble is unnecessarily fragile [vs.r1.p1299r2]

- ¹ Prior revisions of this proposal prohibited macros imported from header units from affecting the set of imports of a module. However, the complexity of the resulting rule – for both users and implementations – was not considered to be justified by the expected benefit for tools wishing to perform dependency extraction, so this rule has been removed.
- ² In this proposal, macros imported from a header unit become available immediately after the import as described in 3.3.3, but unlike in R0, such macros can be expanded prior to later imports in the preamble of a module unit.

3.2.2 P1242R1: Single-file modules [vs.r1.p1242r1]

- ¹ As described in 3.4, this proposal removes the Modules TS “attendant entities” rule. This left a feature vacuum: there was no longer a way to define a module entirely in a single file without module implementation details being available outside the module.

- ² In this proposal, a complete module (with both interface and implementation) can be defined in a single source file by separating the interface from the implementation with a

```
module :private;
```

marker.

3.3 Changes since R0

[vs.r0]

- ¹ This section lists changes to the design of the merged modules proposal since P1103R0.

3.3.1 Namespace export

[vs.r0.namespace]

- ¹ In P1103R0 and in the Modules TS, all namespaces (excluding anonymous namespaces and those nested within them) that are declared in a module interface unit have external linkage and are exported. Following strong EWG direction in Rapperswil, in this document such namespace names are only exported if they are either explicitly exported, or if any name within them is exported. [*Note*: The new approach permits implementation-detail namespace names to be hidden from the interface of a module despite being declared in a module interface unit. — *end note*] [*Example*:

```
export module M;

export namespace A {} // exported
namespace B {        // exported
    export int n;
}
namespace C {        // not exported in this proposal, exported in TS / P1103R0
    int n;
}
```

— *end example*]

3.3.2 Reachability in template instantiations

[vs.r0.templates]

- ¹ Following discussion and direction from Bellevue, we use a path of instantiation rule to guide visibility and reachability of declarations within a template instantiation, based on the relevant rule from the Atom proposal:

Within a template instantiation, the *path of instantiation* is a sequence of locations within the program, starting from the ultimate point of instantiation, via each intervening template instantiation, terminating at the instantiation in question. Names are visible and semantic properties are available within template instantiations if they would be visible or available at any point along the path of instantiation, or (for points outside the current translation unit) would be visible or available at the end of the translation unit containing the relevant point of instantiation.

— P0947R1, 7.1 Templates and two-phase name lookup

This rule permits a template to make use of all declarations that were visible or reachable at each point along its path of instantiation, even if those declarations are not visible or reachable in the template definition context nor the template instantiation context.

- ² [*Example*:

```
export module A;
export template<typename T, typename U> void f(T t, U u) {
    t.f();
}
```

```

module;
struct S { void f(); };
export module B;
import A;
export template<typename U> void g(U u) { S s; f(s, u); }

export module C;
import B;
export template<typename U> void h(const U &u) { g(u); }

import C;
int main() { h(0); }

```

The definition of `struct S` and the declaration of its member `f` are not reachable from the point of instantiation of `f<S, int>`, nor from the template definition. But this code is valid under this proposal, because `S` is reachable from module `B`, which is on the path of instantiation. — *end example*]

- ³ As described above (2.3.3), implementations are permitted to treat additional declarations as reachable even if they would not be reachable on the path of instantiation, if they are transitively imported at the point of instantiation. [*Example*:

```

module M;
struct S;
import C;
// unspecified whether a definition of S is reachable
// here or in the instantiation of h<S>
void q(const S &s) { h(s); }

```

— *end example*]

- ⁴ The same rule applies to the set of names found by ADL: names visible along the path of instantiation are visible to ADL. Internal-linkage declarations within the global module are ignored. In addition, exported declarations in the owning module of each associated type are visible to ADL.

3.3.3 Finding the end of the preamble

[vs.r0.preamble]

- ¹ In R0 of this proposal, the preprocessor was burdened with finding the end of the preamble, and making macros from header units visible at that point. That was problematic both for implementers (as it is a challenging rule to implement) and for users (as code would silently do something different from what was expected, and imports in a preamble would behave differently from imports in non-modular code). This proposal uses a simpler rule: imported macros become visible immediately after the import declaration.

3.4 Changes to the Modules TS

[vs.ts]

- ¹ This section lists the ways in which valid code under the Modules TS would become invalid or change meaning in this merged proposal.
- ² A `module;` introducer is required prior to a global module fragment, as described in P0713R1 and approved by Evolution.
- ³ When an entity is owned by a module and is never exported, but is referenced by an exported part of the module interface, the Modules TS would export the semantic properties associated with the entity at the point of the export. If multiple such exports give the entity different semantics, the program is ill-formed:

```

export module M;
struct S;
export S f(); // S incomplete here
struct S {};
export S g(); // S complete here, error

```

Under the Atom proposal, the semantics of such entities are instead determined their the properties at the end of the module interface unit.

In this merged proposal, the semantics of all entities owned by a module are determined by their properties at the end of the module interface unit (regardless of whether they are exported). [*Note*: The order in which declarations appear within a module interface has no bearing on which semantic properties are exported in this merged proposal. — *end note*]

The Modules TS “attendant entities” rule is removed, because there are no longer any cases where it could apply.

- ⁴ Entities declared within **extern "C"** and **extern "C++"** within a module are no longer owned by that module. It is unclear whether this is a change from the intent of the Modules TS.
- ⁵ Namespace names are exported less often in this proposal, as discussed above.

3.5 Changes relative to the Atom proposal [vs.atom]

- ¹ This section lists the ways in which valid code under the Atom proposal would become invalid or change meaning in this merged proposal.
- ² The merged proposal supports global module fragments, which interferes with the Atom proposal’s goal of making the preamble easy to identify and process with non-compiler tools. However, the benefits of the Atom approach are still available to those who choose not to put code in the global module fragment.
- ³ The identifiers **export** and **module** are taken as keywords by the merged proposal, rather than making them context-sensitive as proposed by the Atom proposal. This follows EWG’s direction on this question from discussion of P0924R0.

Part II

**Wording for applying the merged modules
proposal to the C++20 working draft**

5 Lexical conventions

[lex]

5.1 Separate translation

[lex.separate]

Modify paragraph 5.1/2 as follows

- ² [Note: Previously translated translation units and instantiation units can be preserved individually or in libraries. The separate translation units of a program communicate (6.5) by (for example) calls to functions whose identifiers have external [or module](#) linkage, manipulation of objects whose identifiers have external [or module](#) linkage, or manipulation of data files. Translation units can be separately translated and then later linked to produce an executable program (6.5). — end note]

5.2 Phases of translation

[lex.phases]

Modify bullet 7 of paragraph 5.2/1 as follows:

7. White-space characters separating tokens are no longer significant. Each preprocessing token is converted into a token (5.6). The resulting tokens are syntactically and semantically analyzed and translated as a translation unit. [Note: The process of analyzing and translating the tokens may occasionally result in one token being replaced by a sequence of other tokens (17.2). — end note] [It is implementation-defined whether the sources for module units and header units on which the current translation unit has an interface dependency \(100.1, 100.3\) are required to be available.](#) [Note: Source files, translation units and translated translation units need not necessarily be stored as files, nor need there be any one-to-one correspondence between these entities and any external representation. The description is conceptual only, and does not specify any particular implementation. — end note]

5.4 Preprocessing tokens

[lex.pptoken]

Modify bullet 3 of paragraph 5.4/3 as follows:

Otherwise, the next preprocessing token is the longest sequence of characters that could constitute a preprocessing token, even if that would cause further lexical analysis to fail, except that a *header-name* (5.8) is only formed

- within a `#include` directive (19.2),
- [within a *has-include-expression*, or](#)
- [outside of any preprocessing directive, if applying phase 4 of translation to the sequence of preprocessing tokens produced thus far is valid and results in an *import-seq* \(14.4\).](#)

5.10 Identifiers

[lex.name]

In 5.10, add these two identifiers to Table 4, “Identifiers with special meaning”:

[module](#)
[import](#)

5.11 Keywords

[lex.key]

Modify note in paragraph 5.11/1 as follows:

- ¹ ...
[Note: The ~~export~~ and register keywords ~~are~~ [is](#) unused but ~~are~~ [is](#) reserved for future use. — end note]

6 Basic concepts

[basic]

6.1 Declarations and definitions

[basic.def]

Modify paragraph 6.1/1 as follows:

- ¹ A declaration (Clause 9) may introduce one or more names into a translation unit or redeclare names introduced by previous declarations. If so, the declaration specifies the interpretation and ~~attributes~~ semantic properties of these names. [...]

6.2 One-definition rule

[basic.def.odr]

Change paragraph 6.2/1 as follows:

- ¹ ~~No translation unit shall contain more than one definition of any~~ A variable, function, class type, enumeration type, or template shall not be defined where a prior definition is necessarily reachable (100.6); no diagnostic is required if the prior declaration is in another translation unit.

Modify the end of paragraph 6.2/10 as follows

- ¹⁰ [...] ~~An~~ A definition of an inline function or variable shall be ~~defined~~ reachable in every translation unit in which it is odr-used outside of a discarded statement.

Modify paragraph 6.2/11 as follows

- ¹¹ ~~Exactly one~~ A definition of a class is required to be reachable in a translation unit if in every context in which the class is used in a way that requires the class type to be complete.

Modify opening of paragraph 6.2/12 as follows

- ¹² There can be more than one definition of a class type (Clause 10), enumeration type (9.6), inline function with external linkage (9.1.6), inline variable with external linkage (9.1.6), class template (Clause 12), non-static function template (12.6.6), static data member of a class template (12.6.1.3), member function of a class template (12.6.1.1), or template specialization for which some template parameters are not specified (12.8, 12.6.5) in a program provided that ~~each definition appears in a different translation unit~~ no prior definition is necessarily reachable (100.6) at the point where a definition appears, and provided the definitions satisfy the following requirements. There shall not be more than one definition of an entity that is attached to a named module (100.1); no diagnostic is required unless a prior definition is reachable at a point where a later definition appears. Given such an entity named D defined in more than one translation unit, then

6.3 Scope

[basic.scope]

6.3.6 Namespace scope

[basic.scope.namespace]

Modify paragraph 6.3.6/1 as follows:

- ¹ The declarative region of a *namespace-definition* is its *namespace-body*. Entities declared in a *namespace-body* are said to be members of the namespace, and names introduced by these declarations into the declarative region of the namespace are said to be *member names* of the namespace. A namespace member name has namespace scope. Its potential scope includes its namespace from the name's point of declaration (6.3.2) onwards; and for each *using-directive* (9.7.3) that nominates the member's namespace, the member's potential scope includes that portion of the potential scope of the *using-directive* that follows the member's point of declaration. If a translation unit *Q* is imported into a translation unit *R* (100.3), the potential scope of a name *X* declared with namespace scope in *Q* is extended to include the portion of the corresponding namespace scope in *R* following the first *module-import-declaration* or *module-declaration* in *R* that imports *Q* (directly or indirectly) if

- (1.1) — *X* does not have internal linkage, and
- (1.2) — *X* is declared after the *module-declaration* in *Q* (if any), and
- (1.3) — either *X* is exported or *Q* and *R* are part of the same module.

[*Note: A module-import-declaration imports both the named translation unit(s) and any modules named by exported module-import-declarations within them, recursively. [Example:*

```
// Translation unit #1
export module Q;
export int sq(int i) { return i*i; }

// Translation unit #2
export module R;
export import Q;

// Translation unit #3
import R;
int main() { return sq(9); }      // OK: sq from module Q
```

—end example] —end note]

6.4 Name lookup

[basic.lookup]

Modify paragraph 6.4/1 as follows:

- 1 The name lookup rules apply uniformly to all names (including *typedef-names* (9.1.3), *namespace-names* (9.7), and *class-names* (10.1)) wherever the grammar allows such names in the context discussed by a particular rule. Name lookup associates the use of a name with a set of declarations (6.1) of that name. [...] Only after name lookup, function overload resolution (if applicable) and access checking have succeeded are the **attributes****semantic properties** introduced by the name's declaration and its reachable (100.6) redeclarations used further in the expression processing (Clause 7).

6.4.2 Argument-dependent name lookup

[basic.lookup.argdep]

Modify paragraph 6.4.2/2 as follows:

- 2 For each argument type *T* in the function call, there is a set of zero or more *associated namespaces* and a set of zero or more *associated **classes** entities (other than namespaces)* to be considered. The sets of namespaces and **classes** entities are determined entirely by the types of the function arguments (and the namespace of any template template argument). Typedef names and *using-declarations* used to specify the types do not contribute to this set. The sets of namespaces and **classes** entities are determined in the following way:
 - (2.1) — If *T* is a fundamental type, its associated sets of namespaces and **classes** entities are both empty.
 - (2.2) — If *T* is a class type (including unions), its associated **classes** entities are the class itself; the class of which it is a member, if any; and its direct and indirect base classes. Its associated namespaces are the innermost enclosing namespaces of its associated **classes** entities. Furthermore, if *T* is a class template specialization, its associated namespaces and **classes** entities also include: the namespace and **classes** entities associated with the types of the template arguments provided for template type parameters (excluding template template arguments); the templates used as template template arguments; the namespaces of which any template template arguments are members; and the classes of which any member template used as template template arguments are members. [*Note: Non-type template arguments do not contribute to the set of associated namespaces. —end note*]

- (2.3) — If T is an enumeration type, its associated namespace is the innermost enclosing namespace of its declaration, and its associated entities are T and, if it is a class member, its associated class is the member's class; else it has no associated class.
- (2.4) — If T is a pointer to U or an array of U, its associated namespaces and classes entities are those associated with U.
- (2.5) — If T is a function type, its associated namespaces and classes entities are those associated with the function parameter types and those associated with the return type.
- (2.6) — If T is a pointer to a data member of class X, its associated namespaces and classes entities are those associated with the member type together with those associated with X.

If an associated namespace is an inline namespace (9.7.1), its enclosing namespace is also included in the set. If an associated namespace directly contains inline namespaces, those inline namespaces are also included in the set. In addition, if the argument is the name or address of a set of overloaded functions and/or function templates, its associated classes entities and namespaces are the union of those associated with each of the members of the set, i.e., the classes entities and namespaces associated with its parameter types and return type. Additionally, if the aforementioned set of overloaded functions is named with a *template-id*, its associated classes entities and namespaces also include those of its type *template-arguments* and its template *template-arguments*.

Modify paragraph 6.4.2/4 and add /5 as follows:

- 4 When considering an associated namespace N, the lookup is the same as the lookup performed when ~~the associated namespace N~~ is used as a qualifier (6.4.3.2) except that:
 - (4.1) — Any *using-directives* in ~~the associated namespace N~~ are ignored.
 - (4.2) — Any namespace-scope friend functions or friend function templates (10.7.3) declared in ~~associated~~ classes with reachable definitions in the set of associated entities are visible within their respective namespaces even if they are not visible during an ordinary lookup (9.7.1.2).
 - (4.3) — All names except those of (possibly overloaded) functions and function templates are ignored.
 - (4.4) — Any declaration D in N that is in the interface of a named module M (100.2) is visible if there is an associated entity attached to M with the same innermost enclosing non-inline namespace as D.
 - (4.5) — If the lookup is for a dependent name (12.7.2 [temp.dep], 12.7.4.2), any declaration D in N is visible if D would be visible to qualified name lookup (6.4.3.2) at any point in the instantiation context (100.5) of the lookup, unless D is declared in another translation unit, attached to the global module, and is either discarded (100.4) or has internal linkage.

5 [Example:

```
// TU 1
export module M;
namespace R {
    export struct X {};
    export void f(X);
}
namespace S {
    export void f(X, X);
}

// TU 2
export module N;
import M;
export R::X make();
namespace R { static int g(X); }
```

```

template<typename T, typename U> void apply(T t, U u) {
    f(t, u);
    g(t);
}

// TU 3
module Q;
import N;
namespace S {
    struct Z { template<typename T> operator T(); };
}
void test() {
    auto x = make();    // OK, decltype(x) is R::X in module M
    R::f(x);             // ill-formed: R and R::f are not visible here
    f(x);               // OK, calls R::f from interface of M
    f(x, S::Z());       // ill-formed: S::f in module M not considered
                        // even though S is an associated namespace
    apply(x, S::Z());    // OK, S::f is visible in instantiation context, and
                        // R::g is visible even though it has internal linkage
}

```

—end example]

6.4.3 Qualified name lookup

[basic.lookup.qual]

6.4.3.2 Namespace members

[namespace.qual]

Change in paragraph 2:

- ² For a namespace X and name m , the namespace-qualified lookup set $S(X, m)$ is defined as follows: Let $S'(X, m)$ be the set of all declarations of m in X and the inline namespace set of X (9.7.1) whose potential scope (6.3.6 [basic.scope.namespace]) would include the namespace in which m is declared at the location of the *nested-name-specifier*. If $S'(X, m)$ is not empty, $S(X, m)$ is $S'(X, m)$; otherwise, $S(X, m)$ is the union of $S(N_i, m)$ for all namespaces N_i nominated by *using-directives* in X and its inline namespace set.

6.5 Program and linkage

[basic.link]

Change the definition of *translation-unit* in paragraph 6.5/1 to:

translation-unit:

top-level-declaration-seq_{opt}
global-module-fragment_{opt} *module-declaration* *top-level-declaration-seq_{opt}* *private-module-fragment_{opt}*

private-module-fragment:

module : *private* ; *top-level-declaration-seq_{opt}*

top-level-declaration-seq:

top-level-declaration
top-level-declaration-seq *top-level-declaration*

top-level-declaration:

module-import-declaration
declaration

Add new paragraphs after the grammar:

- ¹ A *private-module-fragment* shall appear only in a primary module interface unit (100.1). A module unit with a *private-module-fragment* shall be the only module unit of its module; no diagnostic is required.
- ² A token sequence beginning with *export_{opt}* *module* or *export_{opt}* *import* and not immediately followed by *::* is never interpreted as the *declaration* of a *top-level-declaration*.

Insert a new bullet between first and second bullet of paragraph 6.5/2:

- (2.1) — When a name has *module linkage*, the entity it denotes can be referred to by names from other scopes of the same module unit (100.1) or from scopes of other module units of that same module.

Modify bullet (3.2) of paragraph 6.5/3 as follows:

- (2.2) — a ~~non-inline~~ variable of non-volatile const-qualified type, unless
 (2.2.1) — it is explicitly declared extern, or
 (2.2.2) — it is inline or exported, or
 (2.2.3) — it was previously declared and the prior declaration did not have internal linkage; or
~~that is neither explicitly declared extern nor previously declared to have external linkage;~~
~~or~~

Modify paragraph 6.5/4 as follows:

- 4 An unnamed namespace or a namespace declared directly or indirectly within an unnamed namespace has internal linkage. All other namespaces have external linkage. A name having namespace scope that has not been given internal linkage above ~~has the same linkage as the enclosing namespace if it~~ and that is the name of
- (4.1) — a variable; or
 (4.2) — a function; or
 (4.3) — a named class (Clause 10), or an unnamed class defined in a typedef declaration in which the class has the typedef name for linkage purposes (9.1.3); or
 (4.4) — a named enumeration (9.6), or an unnamed enumeration defined in a typedef declaration in which the enumeration has the typedef name for linkage purposes (9.1.3); or
 (4.5) — a template;
- has its linkage determined as follows:
- (4.6) — if the enclosing namespace has internal linkage, the name has internal linkage;
 (4.7) — otherwise, if the declaration of the name is attached to a named module (100.1) and is not exported (100.2), the name has module linkage;
 (4.8) — otherwise, the name has external linkage.

Modify 6.5/6 as follows:

- 6 The name of a function declared in block scope and the name of a variable declared by a block scope extern declaration have linkage. If such a declaration is attached to a named module, the program is ill-formed. If there is a visible declaration of an entity with linkage having the same name and type, ignoring entities declared outside the innermost enclosing namespace scope, the block scope declaration declares that same entity and receives the linkage of the previous declaration. If there is more than one such matching entity, the program is ill-formed. Otherwise, if no matching entity is found, the block scope entity receives external linkage.

Modify paragraph 6.5/10 and add /11 as follows:

- 10 Two names that are the same (Clause 6) and that are declared in different scopes shall denote the same variable, function, type, template or namespace if
- (10.1) — both names have external or module linkage and are declared in declarations attached to the same module, or else both names have internal linkage and are declared in the same translation unit; and
 (10.2) — both names refer to members of the same namespace or to members, not by inheritance, of the same class; and

- (10.3) — when both names denote functions, the parameter-type-lists of the functions (9.2.3.5) are identical; and
- (10.4) — when both names denote function templates, the signatures (12.6.6.1) are the same.

If multiple declarations of the same name with external linkage would declare the same entity except that they are attached to different modules, the program is ill-formed; no diagnostic is required. [Note: *using-declarations*, *typedef declarations*, and *alias-declarations* do not declare entities, but merely introduce synonyms. Similarly, *using-directives* do not declare entities. —end note]

- 11 If a declaration would redeclare a reachable declaration attached to a different module, the program is ill-formed. [Example:

```
// "decls.h"
int f();           // #1, attached to the global module
int g();           // #2, attached to the global module

// module interface of M
module;
#include "decls.h"
export module M;
export using ::f;   // OK: does not declare an entity, exports #1
int g();           // error: matches #2, but attached to M
export int h();     // #3
export int k();     // #4

// other translation unit
import M;
static int h();     // error: matches #3
int k();           // error: matches #4
```

—end example] As a consequence of these rules, all declarations of an entity are attached to the same module; the entity is said to be *attached* to that module.

6.8 Program execution

[basic.exec]

6.8.3 Start and termination

[basic.start]

6.8.3.1 main function

[basic.start.main]

Modify paragraph 6.8.3.1/1 as follows:

- 1 A program shall contain a global function called `main` attached to the global module.

Modify paragraph 6.8.3.1/3 as follows:

- 3 ... A program that declares a variable `main` at global scope, or that declares a function `main` at global scope attached to a named module, or that declares the name `main` with C language linkage (in any namespace) is ill-formed.

9 Declarations

[dcl.dcl]

Add new alternatives to *declaration* in paragraph 9/1 as follows

declaration:
block-declaration
nodeclspec-function-declaration
function-definition
template-declaration
explicit-instantiation
explicit-specialization
linkage-specification
namespace-definition
empty-declaration
attribute-declaration
export-declaration

9.1 Specifiers

[dcl.spec]

9.1.6 The inline specifier

[dcl.inline]

Modify paragraph 9.1.6/6 as follows

- 6 ~~An~~ If an inline function or variable ~~is odr-used in a~~ is odr-used in a ~~shall be defined in every~~ translation unit, a definition of it shall be reachable from the end of that translation unit, in which it is odr-used and it shall have exactly the same definition in every such translation unit ~~ease~~ (6.5). [Note: A call to the inline function or a use of the inline variable may be encountered before its definition appears in the translation unit. —end note] If ~~the a~~ definition of a function or variable ~~appears in a translation unit before is reachable at the point of~~ its first declaration as inline, the program is ill-formed. If a function or variable with external or module linkage is declared inline in one translation unit, there it shall be a reachable declared inline declaration in all translation units in which it ~~appears is declared~~; no diagnostic is required. An inline function or variable with external or module linkage shall have the same address in all translation units. [Note: A static local variable in an inline function with external or module linkage always refers to the same object. A type defined within the body of an inline function with external or module linkage is the same type in every translation unit. —end note]

Add a new paragraph 9.1.6/7 as follows:

- 7 An exported inline function or variable shall be defined in the translation unit containing its exported declaration, outside the *private-module-fragment* (if any). [Note: There is no restriction on the linkage (or absence thereof) of entities that the function body of an exported inline function can reference. A constexpr function (9.1.5) is implicitly inline. —end note]

9.1.7 Type specifiers

[dcl.type]

9.1.7.4 The auto specifier

[dcl.spec.auto]

Add a new paragraph before 9.1.7.4/9 (“If the name of an entity with an undeduced placeholder type appears in an expression, the program is ill-formed.”) as follows:

- 9 An exported function with a declared return type that uses a placeholder type shall be defined in the translation unit containing its exported declaration, outside the *private-module-fragment* (if any). [Note: There is no restriction on the linkage of the deduced return type. —end note]

9.7 Namespaces

[basic.namespace]

Add a new paragraph after 9.7/1 as follows:

- ¹ A namespace is an optionally-named declarative region. The name of a namespace can be used to access entities declared in that namespace; that is, the members of the namespace. Unlike other declarative regions, the definition of a namespace can be split over several parts of one or more translation units.

- ² [Note: A namespace name with external linkage is exported if any of its *namespace-definitions* is exported, or if it contains any *export-declarations* (100.2). A namespace is never attached to a module, and never has module linkage even if it is not exported. —end note] [Example:

```
export module M;
namespace N1 {}           // N1 is not exported
export namespace N2 {}    // N2 is exported
namespace N3 { export int n; } // N3 is exported
```

—end example]

100 Modules

[module]

Add a new clause between 9 and 10 titled “**Modules**” with contents as follows:

100.1 Module units and purviews

[module.unit]

module-declaration:

```
exportopt module module-name module-partitionopt attribute-specifier-seqopt ;
```

module-name:

```
module-name-qualifieropt identifier
```

module-partition:

```
: module-name-qualifieropt identifier
```

module-name-qualifier:

```
identifier .
```

```
module-name-qualifier identifier .
```

1 A *module unit* is a translation unit that contains a *module-declaration*. A *named module* is the collection of module units with the same *module-name*. The identifiers `module` and `import` shall not appear as *identifiers* in a *module-name* or *module-partition*. The optional *attribute-specifier-seq* appertains to the *module-declaration*.

2 A *module interface unit* is a module unit whose *module-declaration* contains the `export` keyword; any other module unit is a *module implementation unit*. A named module shall contain exactly one module interface unit with no *module-partition*, known as the *primary module interface unit* of the module; no diagnostic is required.

3 A *module partition* is a module unit whose *module-declaration* contains a *module-partition*. A named module shall not contain multiple module partitions with the same *module-partition*. All module partitions of a module that are module interface units shall be directly or indirectly exported by the primary module interface unit (100.3). No diagnostic is required for a violation of these rules. [Note: Module partitions can be imported only by other module units in the same module. The division of a module into module units is not visible outside the module. —end note]

4 [Example:

```
// TU 1
export module A;
export import :Foo;
export int baz();

// TU 2
export module A:Foo;
import :Internals;
export int foo() { return 2 * (bar() + 1); }

// TU 3
module A:Internals;
int bar();

// TU 4
module A;
import :Internals;
int bar() { return baz() - 10; }
int baz() { return 30; }
```

Module A contains four translation units:

- (4.1) — a primary module interface unit,
- (4.2) — a module partition A:Foo, which is a module interface unit forming part of the interface of module A,
- (4.3) — a module partition A:Internals, which does not contribute to the external interface of module A, and
- (4.4) — a module implementation unit providing a definition of bar and baz, which cannot be imported because it does not have a partition name.

—end example]

5 A *module unit purview* is the sequence of *tokens* starting at the *module-declaration* and extending to the end of the translation unit. The *purview* of a named module M is the set of module unit purviews of M's module units.

6 The *global module* is the collection of all *global-module-fragments* and all translation units that are not module units. Declarations appearing in such a context are said to be in the *purview* of the global module. [Note: The global module has no name, no module interface unit, and is not introduced by any *module-declaration*. —end note]

7 A *module* is either a named module or the global module. A declaration is *attached* to a module as follows:

- (7.1) — If the declaration
 - (7.1.1) — is a replaceable global allocation or deallocation function (21.6.2.1 [new.delete.single], 21.6.2.2 [new.delete.array]), or
 - (7.1.2) — is a *namespace-declaration* with external linkage, or
 - (7.1.3) — appears within a *linkage-specification*,
 it is attached to the global module.
- (7.2) — Otherwise, the declaration is attached to the module in whose purview it appears.

8 A *module-declaration* that contains neither `export` nor a *module-partition* implicitly imports the primary module interface unit of the module as if by a *module-import-declaration*. [Example:

```
// TU 1
module B:Y;                // does not implicitly import B
int y();

// TU 2
export module B;
import :Y;                 // OK, does not create interface dependency cycle
int n = y();

// TU 3
module B:X1;               // does not implicitly import B
int &a = n;                 // error: n not visible here

// TU 4
module B:X2;               // does not implicitly import B
import B;
int &b = n;                 // OK

// TU 5
module B;                  // implicitly imports B
int &c = n;                 // OK
```

—end example]

100.2 Export declaration

[module.interface]

```

export-declaration:
    export declaration
    export { declaration-seqopt }

```

1 An *export-declaration* shall appear only at namespace scope and only in the purview of a module interface unit. An *export-declaration* shall not appear directly or indirectly within an unnamed namespace or a *private-module-fragment*. An *export-declaration* has the declarative effects of its *declaration* or its *declaration-seq* (if any). An *export-declaration* does not establish a scope and its *declaration* or *declaration-seq* shall not contain an *export-declaration*.

2 A declaration is *exported* if it is

- (2.1) — a namespace-scope declaration declared within an *export-declaration*, or
- (2.2) — a *module-import-declaration* declared with the `export` keyword (100.3), or
- (2.3) — a *namespace-definition* that contains an exported declaration, or
- (2.4) — a declaration within a header unit (100.3) that introduces at least one name.

The *interface* of a module *M* is the set of all exported declarations within its purview. [Example:

```

export module M;
namespace A {           // exported
    export int f();      // exported
    int g();             // not exported
}

```

The interface of *M* comprises *A* and *A::f*. —end example]

3 An exported declaration shall declare at least one name. If the declaration is not within a header unit, it shall not declare a name with internal linkage.

4 [Example:

```

// "a.h"
export int x;

// TU 1
module;
#include "a.h"           // error: declaration of x is not in the
                        // purview of a module interface unit

export module M;
export namespace {}      // error: does not introduce any names
export namespace {
    int a1;              // error: export of name with internal linkage
}
namespace {
    export int a2;        // error: export of name with internal linkage
}
export static int b;      // error: b explicitly declared static
export int f();           // OK
export namespace N { }    // OK
export using namespace N; // error: does not declare a name

```

—end example]

5 If the declaration is a *using-declaration* (9.8 [namespace.udecl]) and is not within a header unit, all entities to which all of the *using-declarators* ultimately refer (if any) shall have been introduced with a name having external linkage. [Example:

```

// "b.h"
int f();

// "c.h"
int g();

// TU 1
export module X;
export int h();

// TU 2
module;
#include "b.h"
export module M;
import "c.h";
import X;
export using ::f, ::g, ::h;    // OK
struct S;
export using ::S;              // error: S has module linkage
namespace N {
    export int h();
    static int h(int);         // #1
}
export using N::h;              // error: #1 has internal linkage

```

—end example] [Note: Names introduced by typedef declarations and *alias-declarations* are not so constrained. [Example:

```

export module M;
struct S;
export using T = S;    // OK: exports name T denoting type S

```

—end example] —end note]

6

A redeclaration of an exported declaration of an entity is implicitly exported. An exported redeclaration of a non-exported declaration of an entity is ill-formed. [Example:

```

export module M;
struct S { int n; };
typedef S S;
export typedef S S;    // OK, does not redeclare an entity
export struct S;       // error: exported declaration follows non-exported declaration

```

—end example]

7

A name is *exported* by a module if it is introduced or redeclared by an exported declaration in the purview of that module. [Note: Exported names have either external linkage or no linkage; see 6.5. Namespace-scope names exported by a module are visible to name lookup in any translation unit importing that module; see 6.3.6. Class and enumeration member names are visible to name lookup in any context in which a definition of the type is reachable. —end note] [Example:

```

// Interface unit of M
export module M;
export struct X {
    static void f();
    struct Y { };

```

```

};

namespace {
    struct S { };
}
export void f(S);    // OK
struct T { };
export T id(T);      // OK

export struct A;      // A exported as incomplete

export auto rootFinder(double a) {
    return [=](double x) { return (x + a/x)/2; };
}

export const int n = 5; // OK: n has external linkage

// Implementation unit of M
module M;
struct A {
    int value;
};

// main program
import M;
int main() {
    X::f();           // OK: X is exported and definition of X is reachable
    X::Y y;           // OK: X::Y is exported as a complete type
    auto f = rootFinder(2); // OK
    return A{45}.value; // error: A is incomplete
}

```

—end example]

- 8 [Note: Redefining a name in an *export-declaration* cannot change the linkage of the name (6.5).
[Example:

```

// Interface unit of M
export module M;
static int f();           // #1
export int f();           // error: #1 gives internal linkage
struct S;                 // #2
export struct S;          // error: #2 gives module linkage
namespace {
    namespace N {
        extern int x;      // #3
    }
}
export int N::x;           // error: #3 gives internal linkage

```

—end example] —end note]

- 9 [Note: Declarations in an exported *namespace-definition* or in an exported *linkage-specification* (10.5) are exported and subject to the rules of exported declarations. [Example:

```
export module M;
```

```

export namespace N {
    int x;           // OK
    static_assert(1 == 1); // error: does not declare a name
}

```

—end example] —end note]

100.3 Import declaration

[module.import]

module-import-declaration:

```

exportopt import module-name attribute-specifier-seqopt ;
exportopt import module-partition attribute-specifier-seqopt ;
exportopt import header-name attribute-specifier-seqopt ;

```

- 1 In a module unit, all *module-import-declarations* shall precede all other *top-level-declarations* in the *top-level-declaration-seq* of the *translation-unit* and of the *private-module-fragment* (if any). The optional *attribute-specifier-seq* appertains to the *module-import-declaration*.
- 2 A *module-import-declaration* imports a set of translation units determined as described below. [Note: Namespace-scope names exported by the imported translation units become visible (6.3.6) in the importing translation unit and declarations within the imported translation units become reachable (100.6) in the importing translation unit after the import declaration. —end note]
- 3 A *module-import-declaration* that specifies a *module-name* *M* imports all module interface units of *M*.
- 4 A *module-import-declaration* that specifies a *module-partition* shall only appear after the *module-declaration* in a module unit of some module *M*. Such a declaration imports the so-named module partition of *M*.
- 5 A *module-import-declaration* that specifies a *header-name* *H* imports a synthesized *header unit*, which is a translation unit formed by applying phases 1 to 7 of translation (5.2 [lex.phases]) to the source file or header nominated by *H*, which shall not contain a *module-declaration*. [Note: All declarations within a header unit are implicitly exported (100.2), and are attached to the global module (100.1). —end note] An *importable header* is a member of an implementation-defined set of headers. *H* shall identify an importable header. Two *module-import-declarations* import the same header unit if and only if their *header-names* identify the same header or source file (14.2). [Note: A *module-import-declaration* nominating a *header-name* is also recognized by the preprocessor, and results in macros defined at the end of phase 4 of translation of the header unit being made visible as described in 14.4. —end note] A declaration of a name with internal linkage is permitted within a header unit despite all declarations being implicitly exported (100.2). If such a declaration declares an entity that is odr-used outside the header unit, or by a template instantiation whose point of instantiation is outside the header unit, the program is ill-formed.
- 6 When a *module-import-declaration* imports a translation unit *T*, it also imports all translation units imported by exported *module-import-declarations* in *T*; such translation units are said to be exported by *T*. When a *module-import-declaration* in a module unit imports another module unit of the same module, it also imports all translation units imported by all *module-import-declarations* in that module unit. These rules may in turn lead to the importation of yet more translation units.
- 7 A module implementation unit shall not be exported. [Example:

```

// Translation unit #1
module M:Part;

// Translation unit #2
export module M;
export import :Part; // error: exported partition :Part is an implementation unit

```

—end example]

- 8 A module implementation unit of a module *M* that is not a module partition shall not contain a *module-import-declaration* nominating *M*. [Example:

```
module M;
import M;           // error: cannot import M in its own unit
```

—end example]

- 9 A translation unit *has an interface dependency* on a module unit *U* if it contains a *module-declaration* or *module-import-declaration* that imports *U* or if it has an interface dependency on a module unit that has an interface dependency on *U*. A translation unit shall not have an interface dependency on itself. [Example:

```
// Interface unit of M1
export module M1;
import M2;

// Interface unit of M2
export module M2;
import M3;

// Interface unit of M3
export module M3;
import M1;           // error: cyclic interface dependency M3 -> M1 -> M2 -> M3
```

—end example]

100.4 Global module fragment

[**module.global**]

global-module-fragment:
 module ; *top-level-declaration-seq_{opt}*

- 1 A *global-module-fragment* specifies the contents of the *global module fragment* for a module unit. The global module fragment can be used to provide declarations that are attached to the global module and usable within the module unit. [Note: Prior to phase 4 of translation, only preprocessing directives can appear in the global module fragment (14.3). —end note]

- 2 A declaration *D* is *decl-reachable* from a declaration *S* in the same translation unit if:

- (2.1) — *D* does not declare a function or function template and *S* contains an *id-expression*, *namespace-name*, *type-name*, *template-name*, or *concept-name* naming *D*, or
- (2.2) — *D* declares a function or function template that is named by an expression (6.2 [basic.def.odr]) appearing in *S*, or
- (2.3) — *S* contains an expression *E* of the form


```
postfix-expression ( expression-listopt )
```

 whose *postfix-expression* denotes a dependent name, or for an operator expression whose operator denotes a dependent name, and *D* is found by name lookup for the corresponding name in an expression synthesized from *E* by replacing each type-dependent argument or operand with a value of a placeholder type with no associated namespaces or entities, or
- (2.4) — *S* contains an expression that takes the address of an overloaded function (11.4 [over.over]) whose set of overloads contains *D* and for which the target type is dependent, or
- (2.5) — there exists a declaration *M* that is not a *namespace-definition* for which *M* is decl-reachable from *S* and either
 - (2.5.1) — *D* is decl-reachable from *M*, or
 - (2.5.2) — *D* redeclares the entity declared by *M* or *M* redeclares the entity declared by *D*, and *D* is neither a friend declaration nor a block-scope declaration, or

- (2.5.3) — D declares a namespace N and M is a member of N , or
- (2.5.4) — one of M and D declares a class or class template C and the other declares a member or friend of C , or
- (2.5.5) — one of D and M declares an enumeration E and the other declares an enumerator of E , or
- (2.5.6) — D declares a function or variable and M is declared in D ,³ or
- (2.5.7) — one of M and D declares a template and the other declares a partial or explicit specialization or an implicit or explicit instantiation of that template, or
- (2.5.8) — one of M and D declares a class or enumeration type and the other introduces a typedef name for linkage purposes for that type.

In this determination, it is unspecified

- (2.6) — whether a reference to an *alias-declaration*, typedef declaration, *using-declaration*, or *namespace-alias-declaration* is replaced by the declarations they name prior to this determination,
- (2.7) — whether a *simple-template-id* that does not denote a dependent type and whose *template-name* names an alias template is replaced by its denoted type prior to this determination,
- (2.8) — whether a *decltype-specifier* that does not denote a dependent type is replaced by its denoted type prior to this determination, and
- (2.9) — whether a non-value-dependent constant expression is replaced by the result of constant evaluation prior to this determination.

³ A declaration D in a global module fragment of a module unit is *discarded* if D is not decl-reachable from any *top-level-declaration* in the *top-level-declaration-seq* of the *translation unit*. [Note: A discarded declaration is neither reachable nor visible to name lookup outside the module unit, nor in template instantiations whose points of instantiation (12.7.4.1 [temp.point]) are outside the module unit, even when the instantiation context (100.5) includes the module unit. — end note]

⁴ [Example:

```
const int size = 2;
int ary1[size];           // unspecified whether size is decl-reachable from ary1
constexpr int identity(int x) { return x; }
int ary2[identity(2)];     // unspecified whether identity is decl-reachable from ary2

template<typename> struct S;
template<typename, int> struct S2;
constexpr int g(int);

template<typename T, int N>
S<S2<T, g(N)>> f();        // S, S2, g, and :: are decl-reachable from f

template<int N>
void h() noexcept(g(N) == N); // g and :: are decl-reachable from h
```

—end example]

⁵ [Example:

```
// "foo.h"
namespace N {
    struct X {};
    int d();
    int e();
    inline int f(X, int = d()) { return e(); }
    int g(X);
```

3) A declaration can appear within a *lambda-expression* in the initializer of a variable.

```

    int h(X);
}

// module M interface
module;
#include "foo.h"
export module M;
template<typename T> int use_f() {
    N::X x; // N::X, N, and :: are decl-reachable from use_f
    return f(x, 123); // N::f is decl-reachable from use_f
                    // N::e is indirectly decl-reachable from use_f
                    // because it is decl-reachable from N::f
                    // N::d is decl-reachable from use_f
                    // because it is decl-reachable from N::f
                    // even though it is not used in this call
}
template<typename T> int use_g() {
    N::X x; // N::X, N, and :: are decl-reachable from use_f
    return g((T(), x)); // N::g is not decl-reachable from use_g
}
template<typename T> int use_h() {
    N::X x; // N::X, N, and :: are decl-reachable from use_h
    return h((T(), x)); // N::h is not decl-reachable from use_h, but
                    // N::h is decl-reachable from use_h<int>
}
int k = use_h<int>();
// use_h<int> is decl-reachable from k, so
// N::h is decl-reachable from k

// module M implementation
module M;
int a = use_f<int>(); // OK
int b = use_g<int>(); // error: no viable function for call to g;
                    // g is not decl-reachable from purview of
                    // module M's interface, so is discarded
int c = use_h<int>(); // OK
—end example]

```

100.5 Instantiation context

[module.context]

- 1 The *instantiation context* is a set of points within the program that determines which names are visible to argument-dependent name lookup (6.4.2) and which declarations are reachable (100.6) in the context of a particular declaration or template instantiation.
- 2 During the implicit definition of a defaulted special member function (10.2.3 [special]), the instantiation context is the union of the instantiation context from the definition of the class and the instantiation context of the program construct that resulted in the implicit definition of the special member function.
- 3 During the implicit instantiation of a template whose point of instantiation is specified as that of an enclosing specialization (12.7.4.1 [temp.point]), the instantiation context is the union of the instantiation context of the enclosing specialization and, if the template is defined in a module interface unit of a module *M* and the point of instantiation is not in a module interface unit of *M*, the point at the end of the *top-level-declaration-seq* of the primary module interface unit of *M* (prior to the *private-module-fragment*, if any).
- 4 During the implicit instantiation of a template that is implicitly instantiated because it is referenced from within the implicit definition of a defaulted special member function, the instantiation context is the instantiation context of the defaulted special member function.

5 During the instantiation of any other template specialization, the instantiation context comprises the point of instantiation of the template.

6 In any other case, the instantiation context at a point within the program comprises that point.

7 [Example:

```
// translation unit #1
export module stuff;
export template<typename T, typename U> void foo(T, U u) { auto v = u; }
export template<typename T, typename U> void bar(T, U u) { auto v = *u; }

// translation unit #2
export module M1;
import "defn.h";           // provides struct X {};
import stuff;
export template<typename T> void f(T t) {
    X x;
    foo(t, x);
}

// translation unit #3
export module M2;
import "decl.h";           // provides struct X; (not a definition)
import stuff;
export template<typename T> void g(T t) {
    X *x;
    bar(t, x);
}

// translation unit #4
import M1;
import M2;
void test() {
    f(0);
    g(0);
}
```

The call to `f(0)` is valid; the instantiation context of `foo<int, X>` comprises

- (7.1) — the point at the end of translation unit #1,
- (7.2) — the point at the end of translation unit #2, and
- (7.3) — the point of the call to `f(0)`,

so the definition of `X` is reachable (100.6).

It is unspecified whether the call to `g(0)` is valid: the instantiation context of `bar<int, X>` comprises

- (7.4) — the point at the end of translation unit #1,
- (7.5) — the point at the end of translation unit #3, and
- (7.6) — the point of the call to `g(0)`,

so the definition of `X` is not necessarily reachable, as described in 100.6. —end example]

100.6 Reachability

[module.reach]

- 1 A translation unit *U* is *necessarily reachable* from a point *P* if *U* is a module interface unit on which the translation unit containing *P* has an interface dependency, or the translation unit containing *P* imports *U*, in either case prior to *P* (100.3). [Note: While module interface units are reachable even when they are only transitively imported via a non-exported import declaration, namespace-scope names from such module interface units are not visible to name lookup (6.3.6). —end note]
- 2 All translation units that are necessarily reachable are *reachable*. It is unspecified whether additional translation units on which the point within the program has an interface dependency are considered reachable, and under what circumstances.⁴ [Note: It is advisable to avoid depending on the reachability of any additional translation units in programs intending to be portable. —end note]
- 3 A declaration *D* is *reachable* or *necessarily reachable*, respectively, if, for any point *P* in the instantiation context (100.5),
- (3.1) — *D* appears prior to *P* in the same translation unit, or
 - (3.2) — *D* is not discarded (100.4), appears in a translation unit that is reachable or necessarily reachable from *P*, respectively, and either does not appear within a *private-module-fragment* or appears in a *private-module-fragment* of the module containing *P*.
- [Note: Whether a declaration is exported has no bearing on whether it is reachable. —end note]
- 4 The accumulated properties of all reachable declarations of an entity within a context determine the behavior of the entity within that context. [Note: These reachable semantic properties include type completeness, type definitions, initializers, default arguments of functions or template declarations, attributes, visibility of class or enumeration member names to ordinary lookup, etc. Since default arguments are evaluated in the context of the call expression, the reachable semantic properties of the corresponding parameter types apply in that context. [Example:

```
// translation unit #1
export module M:A;
export struct B;

// translation unit #2
module M:B;
struct B {
    operator int();
};

// translation unit #3
module M:C;
import :A;
B b1;                                // error: no reachable definition of struct B

// translation unit #4
export module M;
export import :A;
import :B;
B b2;
export void f(B b = B());

// translation unit #5
module X;
import M;
```

4) Implementations are therefore not required to prevent the semantic effects of additional translation units involved in the compilation from being observed.

```

B b3;                                // error: no reachable definition of struct B
void g() { f(); }                    // error: no reachable definition of struct B

```

—end example] —end note]

5 [Note: An entity can have reachable declarations even if it is not visible to name lookup. —end note] [Example:

```

export module A;
struct X {};
export using Y = X;

module B;
import A;
Y y;           // OK, definition of X is reachable
X x;           // ill-formed: X not visible to unqualified lookup

```

—end example]

10 Classes

[class]

10.2 Class members

[class.mem]

10.2.10 Bit-fields

[class.bit]

Modify paragraph 10.2.10/1 as follows:

¹ [...] The bit-field ~~attribute~~semantic property is not part of the type of the class member. [...]

11 Overloading

[over]

11.5 Overloaded operators

[over.oper]

11.5.8 User-defined literals

[over.literal]

Modify paragraph 11.5.8/7 as follows:

- ⁷ [Note: Literal operators and literal operator templates are usually invoked implicitly through user-defined literals (5.13.8). However, except for the constraints described above, they are ordinary namespace-scope functions and function templates. In particular, they are looked up like ordinary functions and function templates and they follow the same overload resolution rules. Also, they can be declared `inline` or `constexpr`, they can have internal, [module](#), or external linkage, they can be called explicitly, their addresses can be taken, etc. —end note]

12 Templates

[temp]

Modify paragraph 12/4 as follows:

- 4 A *template-declaration* can appear only as a namespace scope or class scope declaration. Its declaration shall not be an *export-declaration*. In a function template declaration, the last component of the *declarator-id* shall not be a *template-id*. [...]

12.7 Name resolution

[temp.res]

12.7.4 Dependent name resolution

[temp.dep.res]

12.7.4.1 Point of instantiation

[temp.point]

Delete paragraph 12.7.4.1/7:

- 7 ~~The instantiation context of an expression that depends on the template arguments is the set of declarations with external linkage declared prior to the point of instantiation of the template specialization in the same translation unit.~~

Change in paragraph 12.7.4.1/8:

- 8 in addition to the points of instantiation described above,
- (8.1) — for any such specialization that has a point of instantiation within the top-level-declaration-seq of the translation unit, prior to the private-module-fragment (if any), the point after the top-level-declaration-seq of the translation-unit is also considered a point of instantiation, and
- (8.2) — for any such specialization that has a point of instantiation within the private-module-fragment, the end of the translation unit is also considered a point of instantiation.

12.7.4.2 Candidate functions

[temp.dep.candidate]

Modify paragraph 12.7.4.2/1 as follows

- 1 For a function call where the *postfix-expression* is a dependent name, the candidate functions are found using the usual lookup rules from the template definition context (6.4.1 [basic.lookup.unqual], 6.4.2). ~~except that:~~ [Note:
- (1.1) — ~~For the part of the lookup using unqualified name lookup (6.4.1 [basic.lookup.unqual]), only function declarations from the template definition context are found.~~
- (1.2) — For the part of the lookup using associated namespaces (6.4.2), ~~only~~ function declarations found in ~~either the template definition context or~~ the template instantiation context are found by this lookup, as described in 6.4.2.
- end note] If the call would be ill-formed or would find a better match had the lookup within the associated namespaces considered all the function declarations with external linkage introduced in those namespaces in all translation units, not just considering those declarations found in the template definition and template instantiation contexts, then the program has undefined behavior.

Append new paragraphs as follows:

- 2 [Example:

```

// header file "X.h"
namespace Q {
    struct X { };
}

// header file "G.h"
namespace Q {
    void g_impl(X, X);
}

// interface unit of M1
module;
#include "X.h"
#include "G.h"
export module M1;
export template<typename T>
void g(T t) {
    g_impl(t, Q::X{ });    // ADL in definition context finds Q::g_impl, g_impl not discarded
}

// interface unit of M2
module;
#include "X.h"
export module M2;
import M1;
void h(Q::X x) {
    g(x);                  // OK
}

```

—end example]

3

[Example:

```

// interface unit of Std
export module Std;
export template<typename Iter>
void indirect_swap(Iter lhs, Iter rhs)
{
    swap(*lhs, *rhs);    // swap not found by unqualified lookup, can be found only via ADL
}

// interface unit of M
export module M;
import Std;

struct S { /* ... */ };
void swap(S&, S&);    // #1

void f(S* p, S* q)
{
    indirect_swap(p, q);    // finds #1 via ADL in instantiation context
}

```

—end example]

4

[Example:

```

// header file "X.h"

```

```

struct X { /* ... */ };
X operator+(X, X);

// module interface unit of F
export module F;
export template<typename T>
void f(T t) {
    t + t;
}

// module interface unit of M
module;
#include "X.h"
export module M;
import F;
void g(X x) {
    f(x);           // OK: instantiates f from F,
                    // operator+ is visible in instantiation context
}

```

—end example]

5

[Example:

```

// module interface unit of A
export module A;
export template<typename T>
void f(T t) {
    cat(t, t);      // #1
    dog(t, t);      // #2
}

// module interface unit of B
export module B;
import A;
export template<typename T, typename U>
void g(T t, U u) {
    f(t);
}

// "foo.h", not an importable header
struct foo {
    friend int cat(foo, foo);
};
int dog(foo, foo);

// module interface unit of C1
module;
#include "foo.h" // dog not referenced, discarded
export module C1;
import B;
export template<typename T>
void h(T t) {
    g(foo{ }, t);
}

// translation unit

```

```

import C1;
void i() {
    h(0);          // error: dog not found at #2
}

// "bar.h", an importable header
struct bar {
    friend int cat(bar, bar);
};
int dog(bar, bar);

// module interface unit of C2
module;
#include "bar.h" // imports header unit "bar.h"
export module C2;
import B;
export template<typename T>
void j(T t) {
    g(bar{ }, t);
}

// translation unit
import C2;
void k() {
    j(0);          // OK, dog found in instantiation context:
                  // visible at end of module interface unit of C2
}

—end example]

```


14 Preprocessing directives

[cpp]

Modify paragraph 14/5 as follows:

- 5 The implementation can process and skip sections of source files conditionally, include other source files, [import macros from header units](#), and replace macros. These capabilities are called *preprocessing*, because conceptually they occur before translation of the resulting translation unit.

14.1 Conditional inclusion

[cpp.cont]

Modify the grammar before 14.1/1 as follows:

```

header-name-tokens:
    string-literal
    < h-pp-tokens >

has-include-expression:
    __has_include ( < h-char-sequence > )
    __has_include ( " q-char-sequence " )
    __has_include ( header-name )
    __has_include ( string-literal )
    __has_include ( < h-pp-tokens > )
    __has_include ( header-name-tokens )

```

Modify paragraph 14.1/2 as follows:

- 2 A *defined-macro-expression* evaluates to 1 if the identifier is currently defined as a macro name (that is, if it is predefined or if [it has one or more active macro definitions \(14.4\)](#), [for example because](#) it has been the subject of a `#define` preprocessing directive without an intervening `#undef` directive with the same subject identifier), 0 if it is not.

Modify paragraph 14.1/3 as follows:

- 3 The ~~third and fourth forms~~ [second form](#) of *has-include-expression* ~~are~~ [is](#) considered only if ~~neither of the first or second forms matches~~ [form does not match](#), in which case the preprocessing tokens are processed just as in normal text.

14.2 Source file inclusion

[cpp.include]

Add a new paragraph after 14.2/6 as follows:

- 7 If the header identified by the *header-name* denotes an importable header (100.3), the preprocessing directive is instead replaced by the *preprocessing-tokens*
- ```
import header-name ;
```

Add a new subclause 14.3 titled “Global module fragment” as follows:

### 14.3 Global module fragment

[cpp.glob.frag]

```

pp-global-module-fragment:
 module ; pp-balanced-token-seq module

```

- 1 If the first two preprocessing tokens at the start of phase 4 of translation are `module ;`, the result of preprocessing shall begin with a *pp-global-module-fragment* for which all *preprocessing-tokens* in the *pp-balanced-token-seq* were produced directly or indirectly by source file inclusion (14.2), and for which the second module *preprocessing-token* was not produced by source file inclusion or macro replacement (14.3 [cpp.replace]). Otherwise, the first two preprocessing tokens at the end of phase 4 of translation shall not be `module ;`.

Add a new subclause 14.4 titled “**Header units**” as follows:

#### 14.4 Header units

[cpp.module]

```

import-seq:
 top-level-token-seqopt exportopt import

top-level-token-seq:
 any pp-balanced-token-seq ending in ; or }

pp-import:
 import header-name pp-import-suffixopt ;
 import header-name-tokens pp-import-suffixopt ;

pp-import-suffix:
 pp-import-suffix-token
 pp-import-suffix pp-import-suffix-token

pp-import-suffix-token:
 any pp-balanced-token other than ;

pp-balanced-token-seq:
 pp-balanced-token
 pp-balanced-token-seq pp-balanced-token

pp-balanced-token:
 pp-ldelim pp-balanced-token-seqopt pp-rdelim
 any preprocessing-token other than a pp-ldelim or pp-rdelim

pp-ldelim: one of
 ([{ <: <%

pp-rdelim: one of
)] } :> %>

```

- 1 A sequence of *preprocessing-tokens* matching the form of a *pp-import* instructs the preprocessor to import macros from the header unit (100.3) denoted by the *header-name*. A *pp-import* is only recognized when the sequence of tokens produced by phase 4 of translation up to the *import* token forms an *import-seq*, and the *import* token is not within the *header-name-tokens* or *pp-import-suffix* of another *pp-import*. The *;* *preprocessing-token* terminating a *pp-import* shall not have been produced by macro replacement (14.3 [cpp.replace]). The *point of macro import* for a *pp-import* is immediately after the *;* terminating the *pp-import*.
- 2 In the second form of *pp-import*, a *header-name* token is formed as if the *header-name-tokens* were the *pp-tokens* of a *#include* directive. The *header-name-tokens* are replaced by the *header-name* token. [ *Note:* This ensures that imports are treated consistently by the preprocessor and later phases of translation. — *end note* ]
- 3 Each *#define* directive encountered when preprocessing each translation unit in a program results in a distinct *macro definition*. Importing macros from a header unit makes macro definitions from a translation unit visible in other translation units. Each macro definition has at most one point of definition in each translation unit and at most one point of undefinition, as follows:
  - (3.1) — The *point of definition* of a macro definition within a translation unit is the point at which its *#define* directive occurs (in the translation unit containing the *#define* directive), or, if the macro name is not lexically identical to a keyword (5.11 [lex.key]) or to the *identifiers* *module* or *import*, the first point of macro import of a translation unit containing a point of definition for the macro definition, if any (in any other translation unit).
  - (3.2) — The *point of undefinition* of a macro definition within a translation unit is the first point at which a *#undef* directive naming the macro occurs after its point of definition, or the first point of macro import of a translation unit containing a point of undefinition for the macro definition, whichever (if any) occurs first.

4 A macro directive is *active* at a source location if it has a point of definition in that translation unit preceding the location, and does not have a point of undefinition in that translation unit preceding the location.

5 If a macro would be replaced or redefined, and multiple macro definitions are active for that macro name, the active macro definitions shall all be valid redefinitions of the same macro (14.3 [cpp.replace]). [Note: The relative order of *pp-imports* has no bearing on whether a particular macro definition is active. —end note]

6 [Example:

```
// translation unit a.h
#define X 123 // #1
#define Y 45 // #2
#define Z a // #3
#undef X // point of undefinition of #1 in a.h

// translation unit b.h
import "a.h"; // point of definition of #1, #2, and #3, point of undefinition of #1 in b.h
#define X 456 // OK, #1 is not active
#define Y 6 // error: #2 is active

// translation unit c.h
#define Y 45 // #4
#define Z c // #5

// translation unit d.h
import "a.h"; // point of definition of #1, #2, and #3, point of undefinition of #1 in d.h
import "c.h"; // point of definition of #4 and #5 in d.h
int a = Y; // OK, active macro definitions #2 and #3 are valid redefinitions
int c = Z; // error: active macro definitions #2 and #3 are not valid redefinitions of Z
```

—end example]

# Annex C Compatibility

[diff]

## C.5 C++ and ISO C++ 2017

[diff.cpp17]

### C.5.1 Clause 5: lexical conventions

[diff.cpp17.lex]

Add new entry as follows:

**Affected subclauses:** 5.8

**Change:** *header-name* tokens are formed in more contexts.

**Rationale:** Required for new features.

**Effect on original feature:** When the identifier `import` is followed by a `<` character, a *header-name* token may be formed. [Example:

```
template<typename> class import {};
import<int> f(); // ill-formed; previously well-formed
::import<int> g(); // OK
```

—end example]

### C.5.3 Clause 9: declarations

[diff.cpp17.dcl]

Add new entry as follows:

**Affected subclauses:** 9.11.1, 9.11.3

**Change:** New identifiers with special meaning.

**Rationale:** Required for new features.

**Effect on original feature:** Top-level declarations beginning with `module` or `import` may either be ill-formed or interpreted differently in this International Standard. [Example:

```
class module;
module *m1; // ill-formed; previously well-formed
::module *m2; // OK

class import {};
import j1; // was variable declaration; now import-declaration
::import j2; // variable declaration
```

—end example]